BASEL III Market Risk Modelling Using Monte Carlo VaR

Regulatory Framework
- Basel III
- BCBS Guidelines
- Internal Model Validation

Model Limitations
- Linear assumption
- Normality assumption
- Tail Risk underestimation

Factor Diagnostics
- R^2 Model Fit
- p-value significance
- Variance Inflation Factor (VIF)

Factor-Based Monte Carlo VaR
CAPM-Style Model
XLK, MTUM, VLUE
Multivariate Normal

Return-Based Monte Carlo VaR
- Normal Distribution
- Mean & Volatility
- 95% 1-Day VaR

Probability Space:
Ω : Outcomes
E : Events
P : Probability Measure
X : Ω → R

Random Variables

VaR Methodologies

Normal Var:
$VaR_\alpha = -(\mu + z_\alpha \sigma)$

Uses mean & volatility

Simulated Portfolio Returns
$R^* = \beta^T f^*$

Definition of VaR
$VaR\alpha = \inf\{x : FL(x) \geq \alpha\}$

Confidence level:
$\alpha = 95\%$

$F_t \sim N(\mu_F, \Sigma_F)$

Monte Carlo Simulation
- Random Sampling
- 10,000 Simulations
- Loss Distribution
- Tail Risk (5%)

F_1t : Tech Factor Return

F_2t : Momentum Factor Retur

F_3t :Value Factor Return

Systematic Risk factor returns

are treated as random variables to capture uncertainty in market

Portfolio Returns

are generated via Monte Carlo simulation to estimate the loss

Simulated (future) portfolio returns

$\varepsilon : \Omega \to R$

Idiosyncratic Risk

$R_t = \beta^T F_t + \varepsilon_t$

Expanded form

$R_t = \beta_1 F_{1,t} + \beta_2 F_{2,t} + \beta_3 F_{3,t} + \varepsilon_t$

$L_t = -R_t$

Portfolio Loss

Generates empirical loss distribution

CDF OF LOSS
$FL(x) = P(L \leq x) = \Phi((x - \mu_l) / \sigma_l)$
$\Phi(z) = \int$ from $-\infty$ to z $[1 / \sqrt{(2\pi)}] e^{(-u^2 / 2)} du$

PDF:
$f_L(x) = 1 / sqrt(2\pi\sigma_L^2) \cdot exp(- (x - \mu_L)^2 / (2\sigma_L^2))$

Portfolio Weights
AAPL 30%
MSFT 25%
NVDA 20%
GOOG 15%
AMD 10%

Defines Portfolio for

Portfolio Construction
- AAPL, MSFT, NVDA
- GOOG, AMD
- Fixed Weights
- Tech Equity Portfolio

computes return for

Return Calculation
- Adjusted Close Prices
- Log Returns
- Portfolio Returns

Train-Test Split
- 80% Training Data
- 20% Testing Data

Depends On

parametrized by

Statistical Estimation
Mean
Volatility
Covariance Matrix

Factor Moments:
$\mu_F = E[F_t], \Sigma_F = Cov(F_t)$

Variance Decomposition
Systematic Risk
Idiosyncratic Risk

Validates

Backtesting & Validation

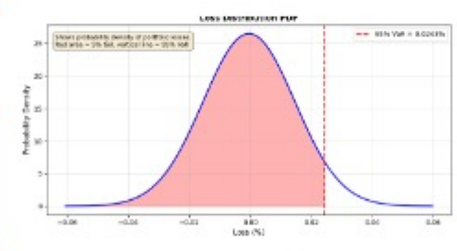
System Pipeline

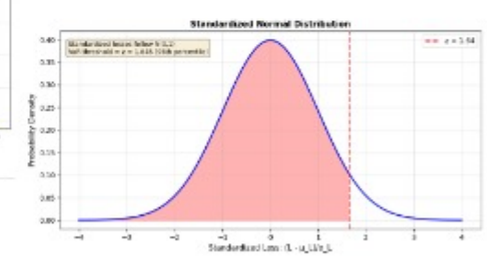
Distributional Assumptions:
- Factor returns vector:
$F_t \sim N(mu_F, Sigma_F)$
- Mean of factors:
$mu_F = E[F_t]$
- Covariance matrix of factors:
$Sigma_F = Cov(F_t) = E[(F_t - mu_F)(F_t - mu_F)^T]$
- Distribution of Portfolio Return: $R_t = \beta^T F_t + \varepsilon_t$
$R_t \sim N(\beta^T mu_F , \beta^T Sigma_F \beta + sigma_epsilon^2)$
- Mean of portfolio return: $mu_R = \beta^T mu_F$
- Variance of portfolio return: $sigma_R^2 = \beta^T Sigma_F \beta + sigma_epsilon^2$

Exception Rate Analysis
Compare predicted VaR with realized portfolio loss

- Historical Backtesting
- Rolling Backtesting (250 Days)
- VaR Exceptions

CSV Market Data
Train Test Split
Portfolio Construction
Loss Distribution
Backtesting



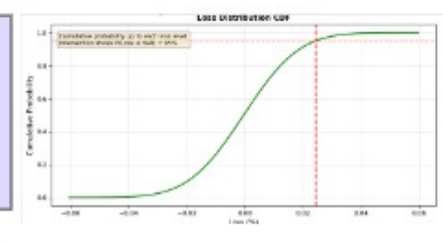


Is Evaluated Using

Basel Traffic Light
- Green (0–4)
- Yellow (5–9)
- Red (10+)

Model Evaluation
Strengths:
- Interpretable
- Efficient

Limitations:
- Gaussian assumption
- Tail risk underestimation



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